Topic 2 Explained Simply

Speech, sucking habits and tongue thrust · the three papers in plain words

🧩 The one question behind all three papers

Do the things we do with our mouths (talking, sucking, swallowing) shape the teeth, and do the teeth shape how we talk? The short answer: **much less than people think**, with one big exception. A force that acts **for many hours a day, for years**, like a thumb or a pacifier, or a tongue that *rests* forward, can move teeth. A force that lasts a second, like a swallow or a spoken sound, cannot.

🗣️ Paper 1 · Johnson and Sandy 1999: Do crooked teeth cause speech problems?

What it is: a review. The authors went through the main studies (no systematic search) and asked what they really show.

The simple version: the tongue is a brilliant adapter. Many people with crooked teeth speak perfectly well, because the tongue quietly finds another way to make each sound. So malocclusion and speech problems are **only loosely linked**.

Where a link does show up:

- **Anterior open bite and lisping.** If the front teeth don't meet, the tongue slips into the gap on /s/. This is the most consistent finding.
- **Class III**, a **big overjet** and **gaps between upper front teeth** also show some link.
- But **worse teeth do not mean worse speech**. A mild open bite can lisp and a severe one can sound normal.

Does fixing the teeth fix the speech? Sometimes it helps, but there is **no proof**. In one small study, orthodontics alone took patients from about 6 faulty sounds to about 1.5, yet the hardest sounds, /s/ /z/ /sh/, stayed wrong. Jaw surgery results are mixed too. So the rule is: **fix teeth for the teeth, and send speech problems to a speech therapist.**

Bloomer's formula, the line to remember: what matters is not the structure alone but whether the person **adapts**. Abnormal teeth with good adaptation give normal speech; normal teeth with poor function still give abnormal speech.

🍼 Paper 2 · Warren and Bishara 2002: Do pacifiers and thumbs damage the bite?

What it is: a real study. They followed **372 Iowa children from birth to age 5**, asked parents about feeding and habits every few months, then took models of the teeth.

What they found, in plain words:

- **Breast-feeding** for longer or shorter made **no difference** to the teeth, at least in children who hardly used a pacifier or thumb.
- **Pacifier for more than about 2 years:** more **posterior crossbite**. The lower arch gets wider and the upper doesn't keep up. With a pacifier for 4 years or more, about **4 in 10** children had a crossbite.
- **Thumb or finger beyond about 3–4 years:** more **open bite** and a **big overjet**. The upper incisors are pushed forward and the upper arch becomes narrower and deeper from front to back. With a thumb habit for 4 years or more, about **4 in 10** had an open bite and about **4 in 10** an overjet of 4 mm or more. The open bite is often **lopsided**, on the side of the hand the child uses.
- **Some of the damage stayed even after the habit stopped**, at least up to age 5.

Why it matters: the official advice at the time said not to worry until the early mixed dentition, around age 6. This paper says that is **too late**: the trouble starts after **2–3 years**.

What is what: *nutritive sucking* means feeding (breast or bottle); *nonnutritive sucking* means comfort sucking (pacifier, thumb, finger).

👅 Paper 3 · Proffit and Mason 1975: Should we treat tongue thrust?

What it is: a review by Proffit himself, using his own measurements of tongue and lip pressure.

The simple version: babies swallow with the **tongue pushed forward** between the gums. Adults swallow with the **teeth together** and the **tongue tip on the palate** behind the front teeth. Children switch over **somewhere between 2 and 12**. A child still pushing the tongue forward is usually **just late, not broken**.

Why young children push the tongue forward: the tongue is nearly full size by age 8 but the jaw is still small, and big tonsils and adenoids crowd the throat. The tongue comes forward to keep the **airway** open. At puberty the jaw grows, the tonsils shrink and there is more room, so the tongue moves back by itself. **Up to 80 %** of children with tongue thrust and open bite at 8 improve by 12 with no treatment.

Does tongue thrust cause open bite? Mostly **no**:

- About half of 6-year-olds tongue-thrust, but only about **6 %** of children have an open bite.
- Proffit measured the pressures, and swallowing pressure does not match tooth position: Australian Aborigines, with **bigger** arches, had **lower** tongue pressure than American whites. What matters is where the tongue **rests** all day. Teeth erupt with a force of only about **5 g**, so a tongue resting forward may stop the front teeth coming down (never shown directly).

What to do:

- Tongue thrust with no speech or tooth problem: **do nothing**.
- With a speech problem: **speech therapy**, not swallowing exercises.
- With a malocclusion: **fix the malocclusion first**; the thrust usually goes away.
- Still there at **puberty**: train the **resting** tongue position, **together with braces**.
- Never do swallowing exercises if the **airway is small**.

📖 What is what

Term	Meaning
Articulation	Shaping sounds with the tongue, lips and teeth

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Lisp (sigmatism)	Faulty /s/: *interdental* = tongue between the teeth; *lateral* = air escapes at the sides
Sibilants, fricatives	Hissing sounds made through a narrow gap: /s/ /z/ /sh/ /zh/
Mesial occlusion	Class III; *postnormal* = Class II
Risk ratio ×3.4	3.4 times as likely as in someone without that trait
Myofunctional therapy	Exercises to retrain the tongue and lips
Deglutition	Swallowing; *infantile* or *visceral* swallow = the baby pattern
Faucial isthmus	The opening at the back of the mouth into the throat, narrowed by big tonsils
Equilibrium theory	Teeth sit where the forces around them balance; **how long** a force lasts matters more than how hard it is

🎯 If a consultant asks...

- **"Should I refer a lisping child with open bite for orthodontics to fix the speech?"** Not for the speech alone. Johnson and Sandy found no proof that moving teeth fixes articulation; refer to a speech therapist.
- **"When should a thumb habit stop?"** By about 2–3 years, not the early mixed dentition. Warren and Bishara showed open bite and overjet climbing after 36 months, and the changes persisting at age 5.
- **"Is tongue thrust the cause of this open bite?"** Usually it is an adaptation or a delayed stage. Look at the resting tongue posture and the airway; Proffit found swallowing pressures do not explain tooth position.
- **"What is weak about these papers?"** Paper 1 is a narrative review of old, mostly cross-sectional studies; paper 2 relies on parents' reports and stops at age 5; paper 3 is a 1975 narrative review of the authors' pressure studies and clinical experience, with no trial testing its recommendations.